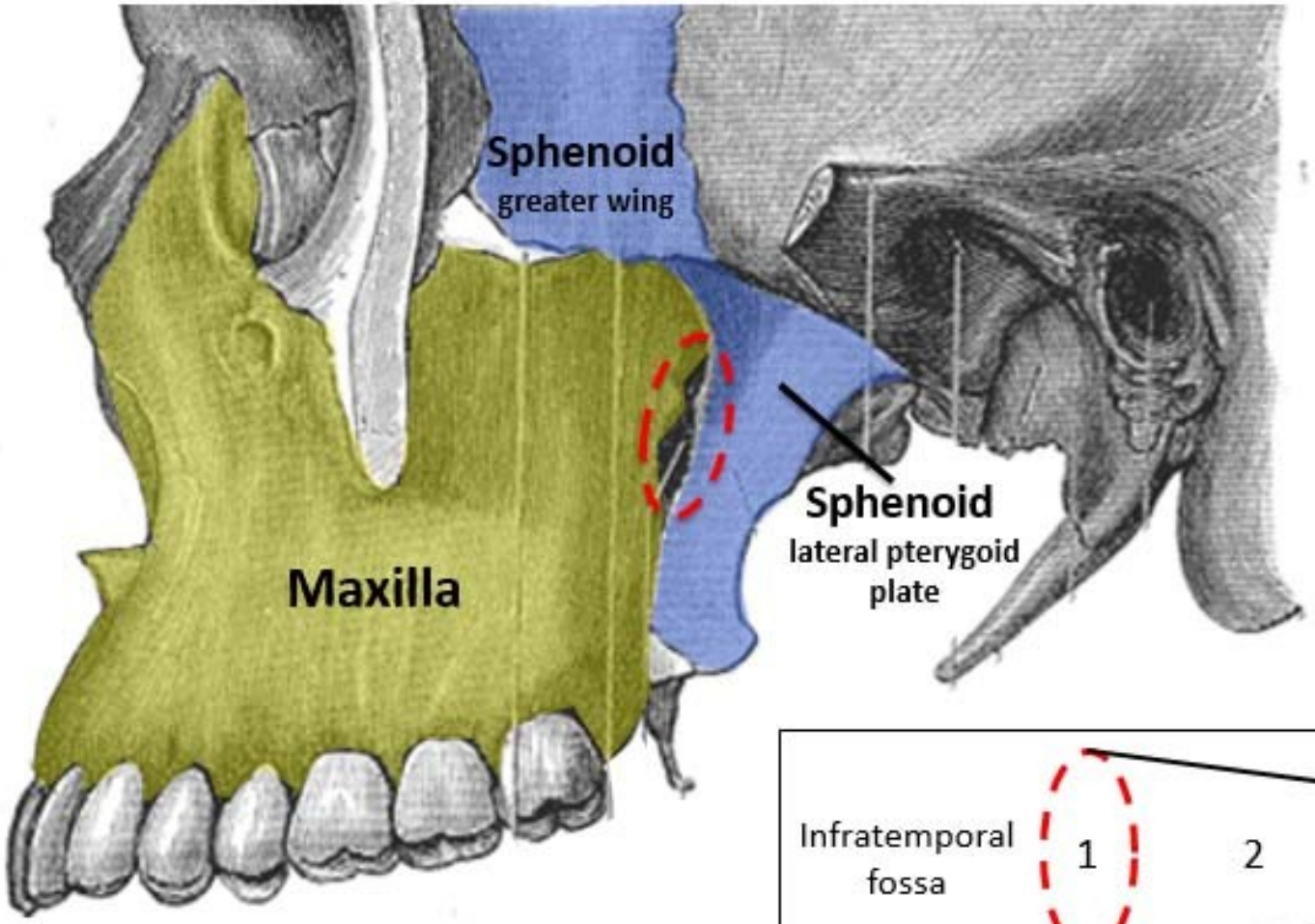
PERYGOPALATINE FOSSA

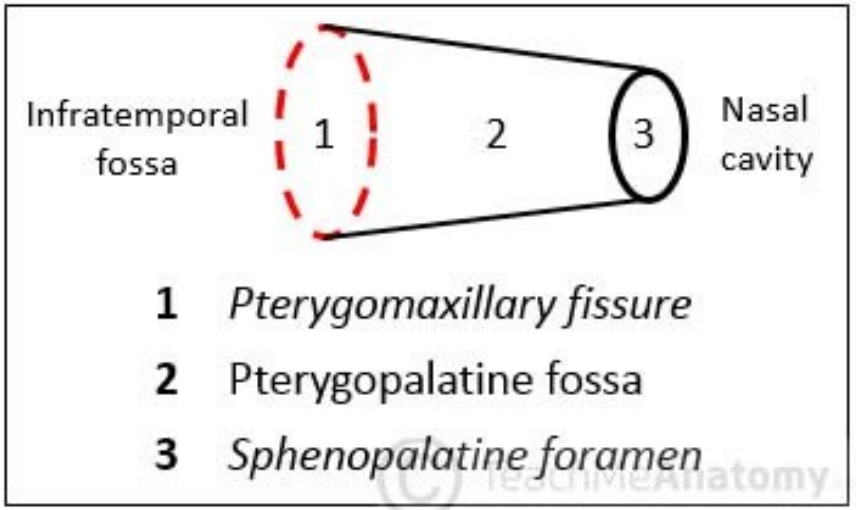
Pterygopalatine fossa

The pterygopalatine fossa is a small space

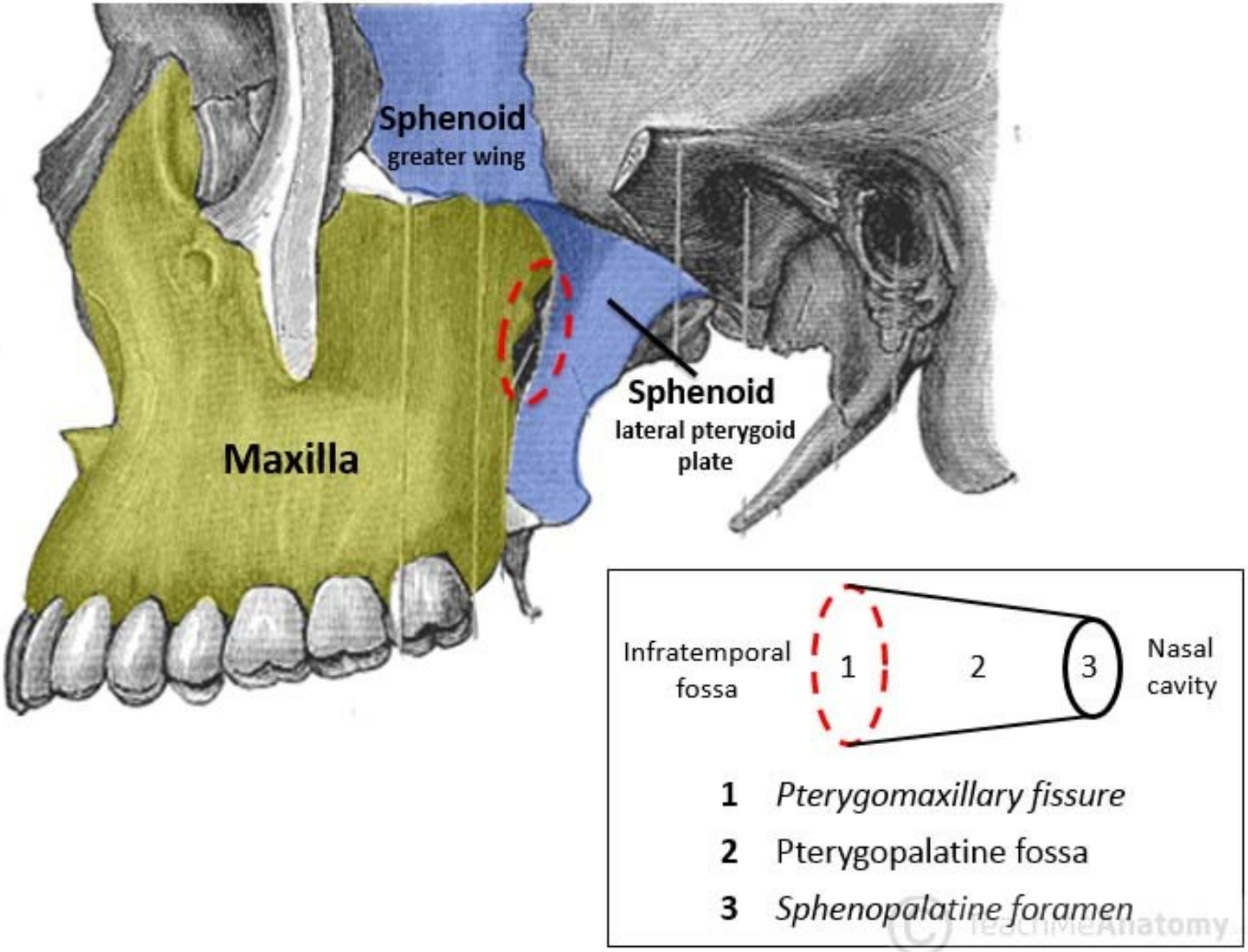
which lies between the body of the maxilla and the

root of the pterygoid process (of the sphenoid bone) lateral to the perpendicular plate of the palatine bone


Sphenoid
greater wing
Sphenoid
lateral pterygoid
plate
Maxilla


Infratemporal
fossa
1
2
3
Nasal
cavity
1 Pterygomaxillary fissure
2 Pterygopalatine fossa
3 Sphenopalatine foramen

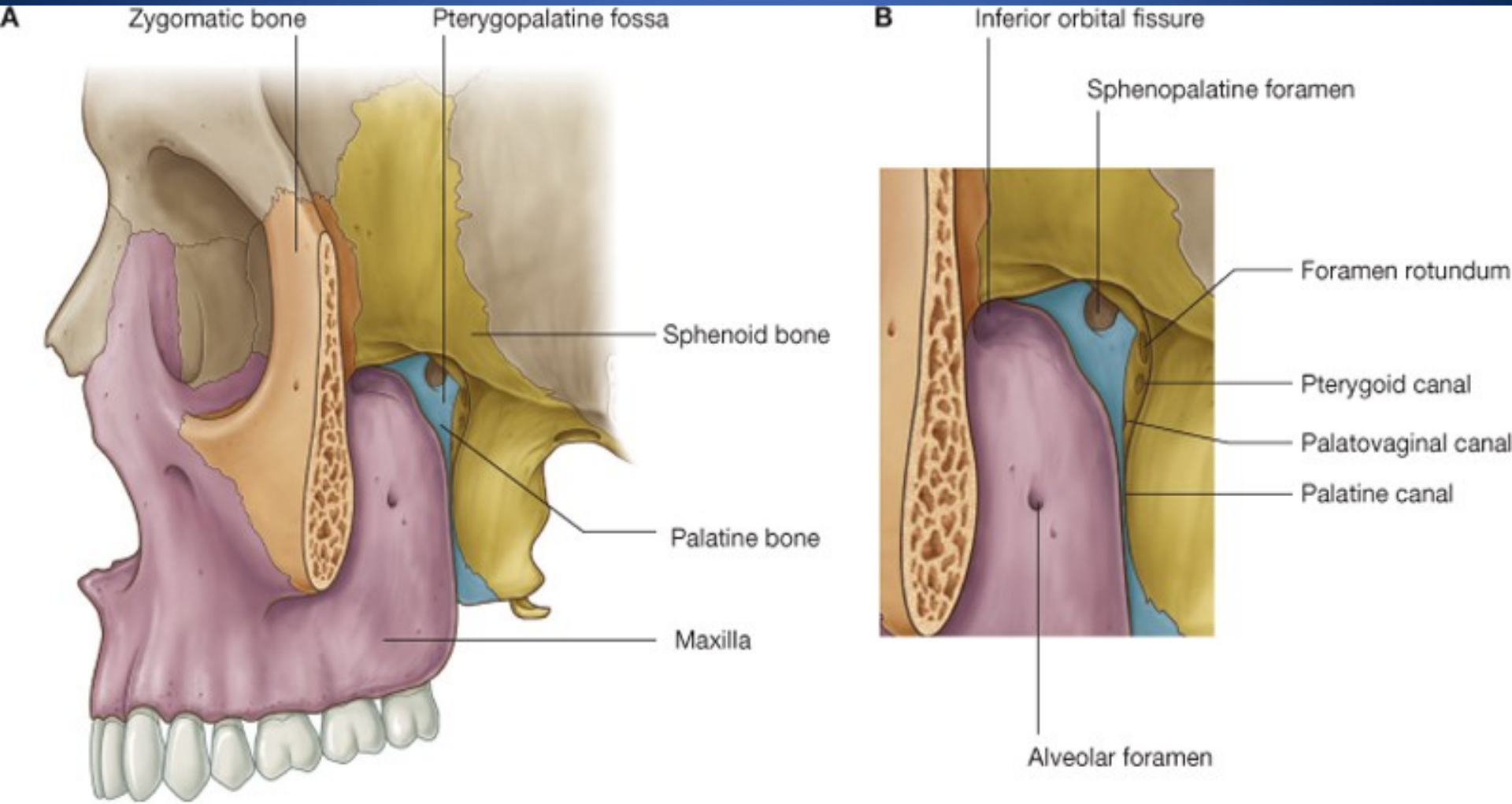
- The **pterygopalatine fossa** is a bilateral, cone-shaped depression extending deep from the infratemporal fossa all the way to the nasal cavity via the sphenopalatine foramen.
- It is located between the maxilla, sphenoid and **palatine** bones, and communicates with other regions of the skull and facial skeleton via several canals and foramina. Its small volume combined with the numerous structures that pass through makes this a complex region.


Sphenoid
greater wing
Sphenoid
lateral pterygoid
plate
Maxilla
Infratemporal
fossa
1
2
3
Nasal
cavity
1 Pterygomaxillary fissure
2 Pterygopalatine fossa
3 Sphenopalatine foramen

Borders

- The borders of the pterygopalatine fossa are formed by the palatine, maxilla and sphenoid bones:
- **Anterior**: Posterior wall of the maxillary sinus.
- **Posterior**: Pterygoid process of the sphenoid bone.
- **Inferior**: Palatine bone and palatine canals.
- **Superior**: Inferior orbital fissure of the eye.
- **Medial**: Perpendicular plate of the palatine bone
- **Lateral**: Pterygomaxillary fissure

BOUNDARIES OF FOSSA



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OPENINGS AT FOSSA

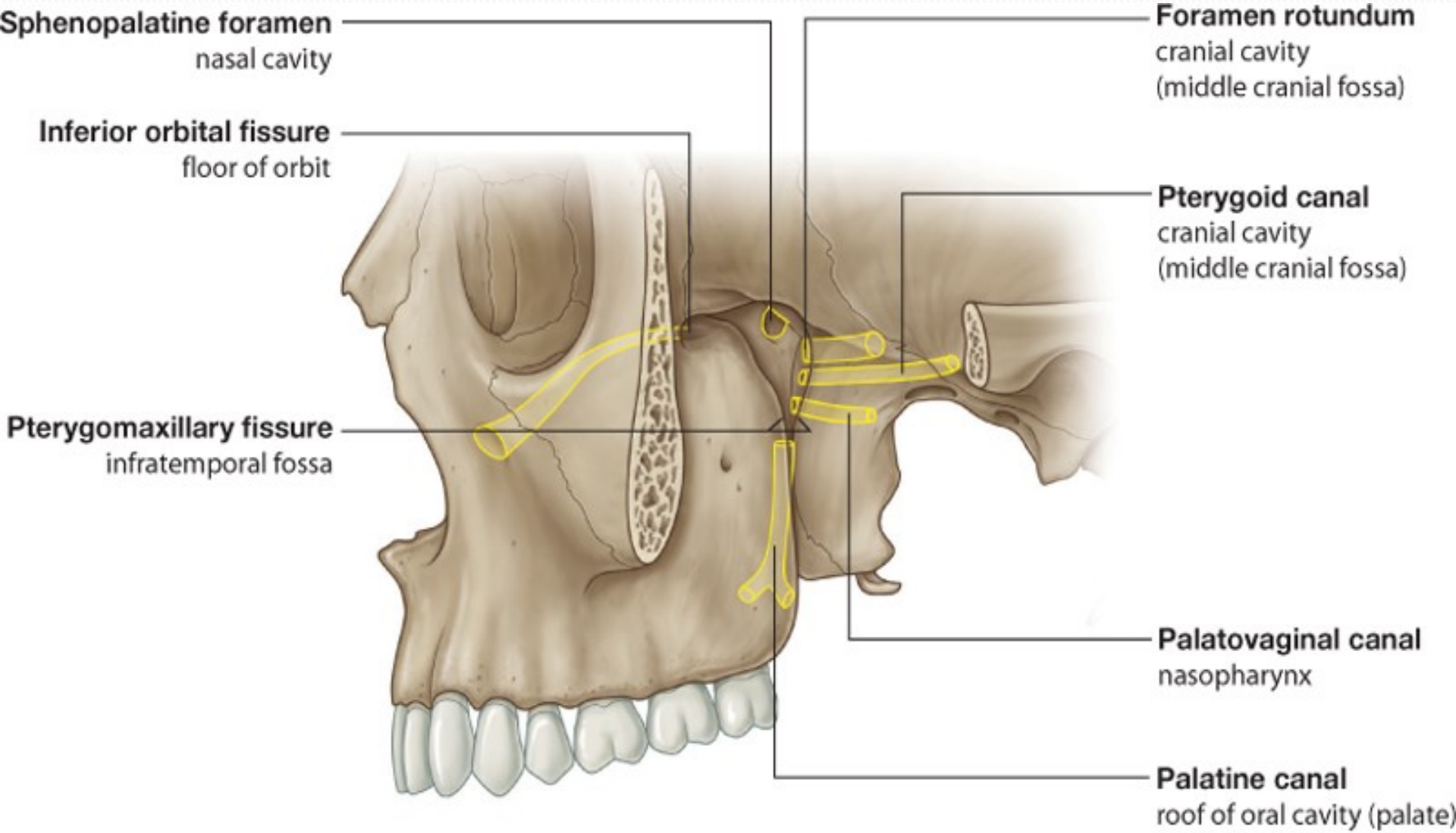
- There are seven openings (also known as foramina) that connect the pterygopalatine fossa with the orbit, nasal and oral cavities, middle cranial fossa and infratemporal fossa. The openings transmit blood vessels and nerves between these regions.

OPENINGS AT FOSSA

1. **Pterygomaxillary fissure**
2. **Foramen rotundum**
3. **Pterygoid canal**
4. **Pharyngeal canal**
5. **Inferior orbital fissure**
6. **Greater palatine canal**
7. **Sphenopalatine foramen**

Pterygomaxillary Fissure

- The pterygomaxillary fissure connects the **infratemporal fossa** with the pterygopalatine fossa (see figure 1). It transmits two neurovascular structures:
- **Posterior superior alveolar nerve** – a branch of the maxillary nerve. It exits through the fissure into the infratemporal fossa, where it goes on to supply the maxillary molars.
- **Terminal part of the maxillary artery** – enters the pterygopalatine fossa via the fissure.

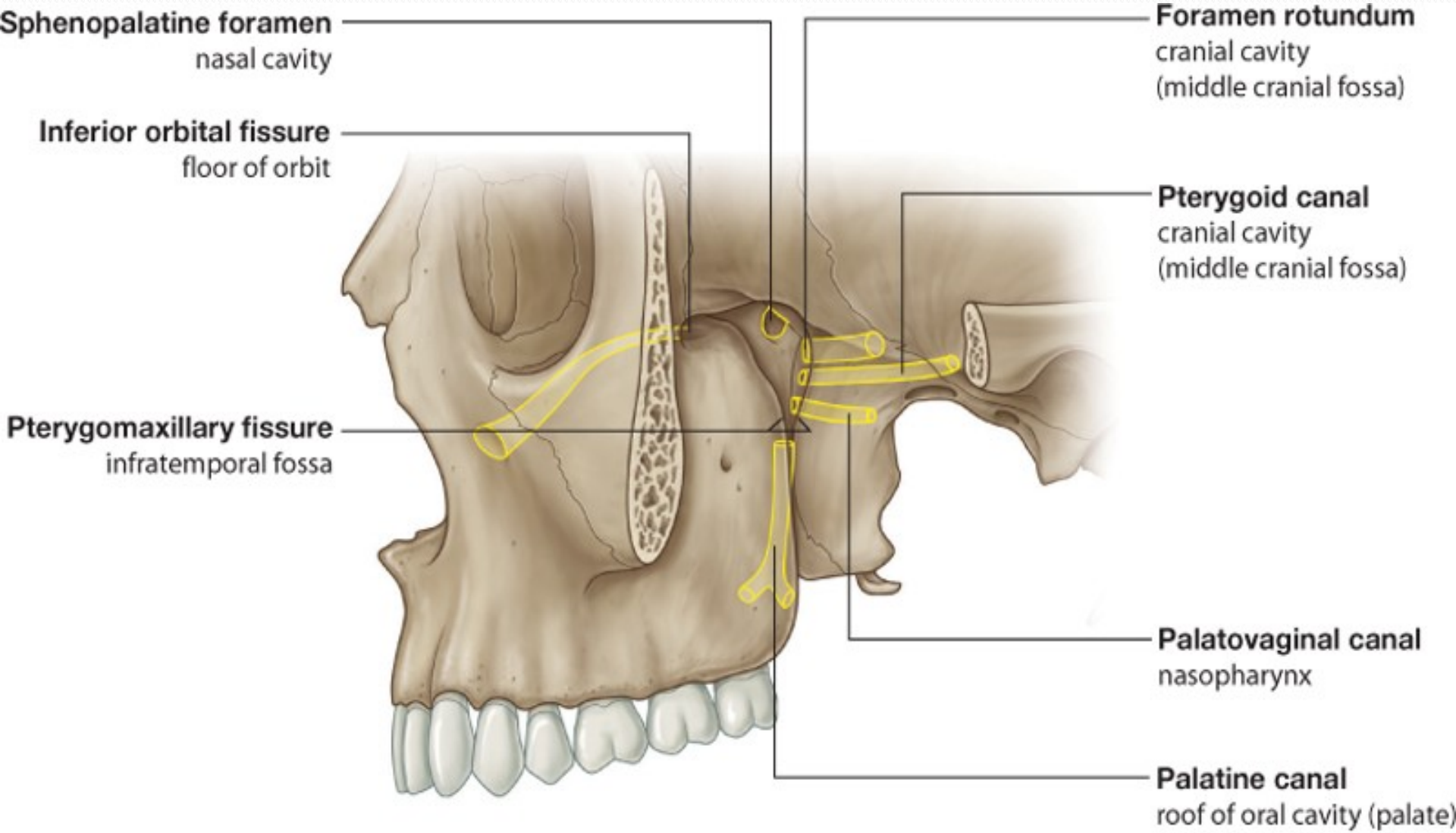


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Foramen Rotundum

- The foramen rotundum connects the pterygopalatine fossa to the **middle cranial fossa**. It is one of three openings in the posterior boundary of the pterygopalatine fossa. It conducts a single structure, the **maxillary nerve.**

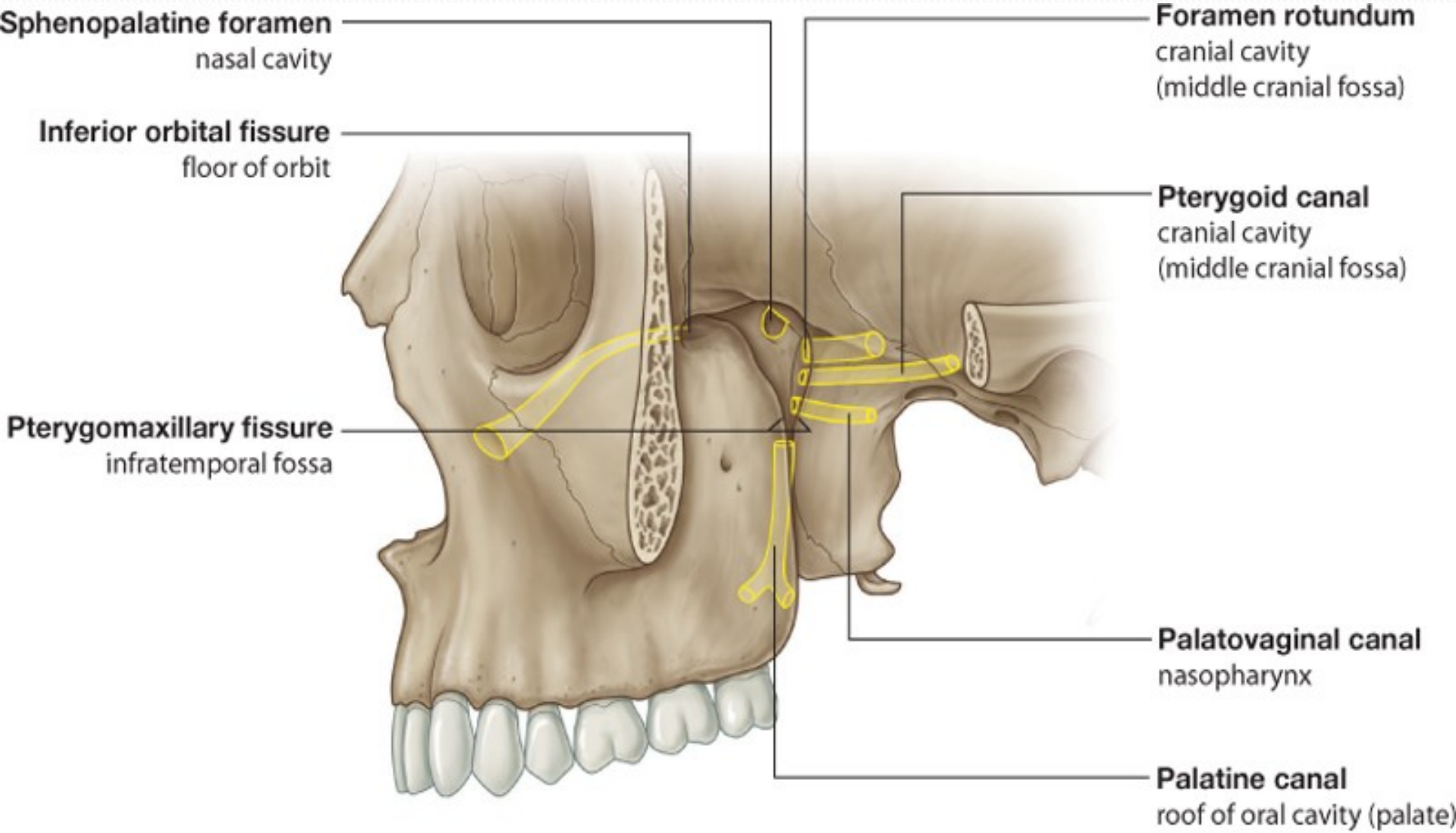


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Pterygoid and Pharyngeal Canals

- These two canals, along with the foramen rotundum, are the three openings in the posterior wall of the pterygopalatine fossa:
- **Pterygoid canal** – runs from the middle cranial fossa and through the medial pterygoid plate. It carries the nerve, artery and vein of the pterygoid canal.
- **Pharyngeal(Palatovaginal canal) canal** – communicates with the nasopharynx. It carries the pharyngeal branches of the maxillary nerve and artery.

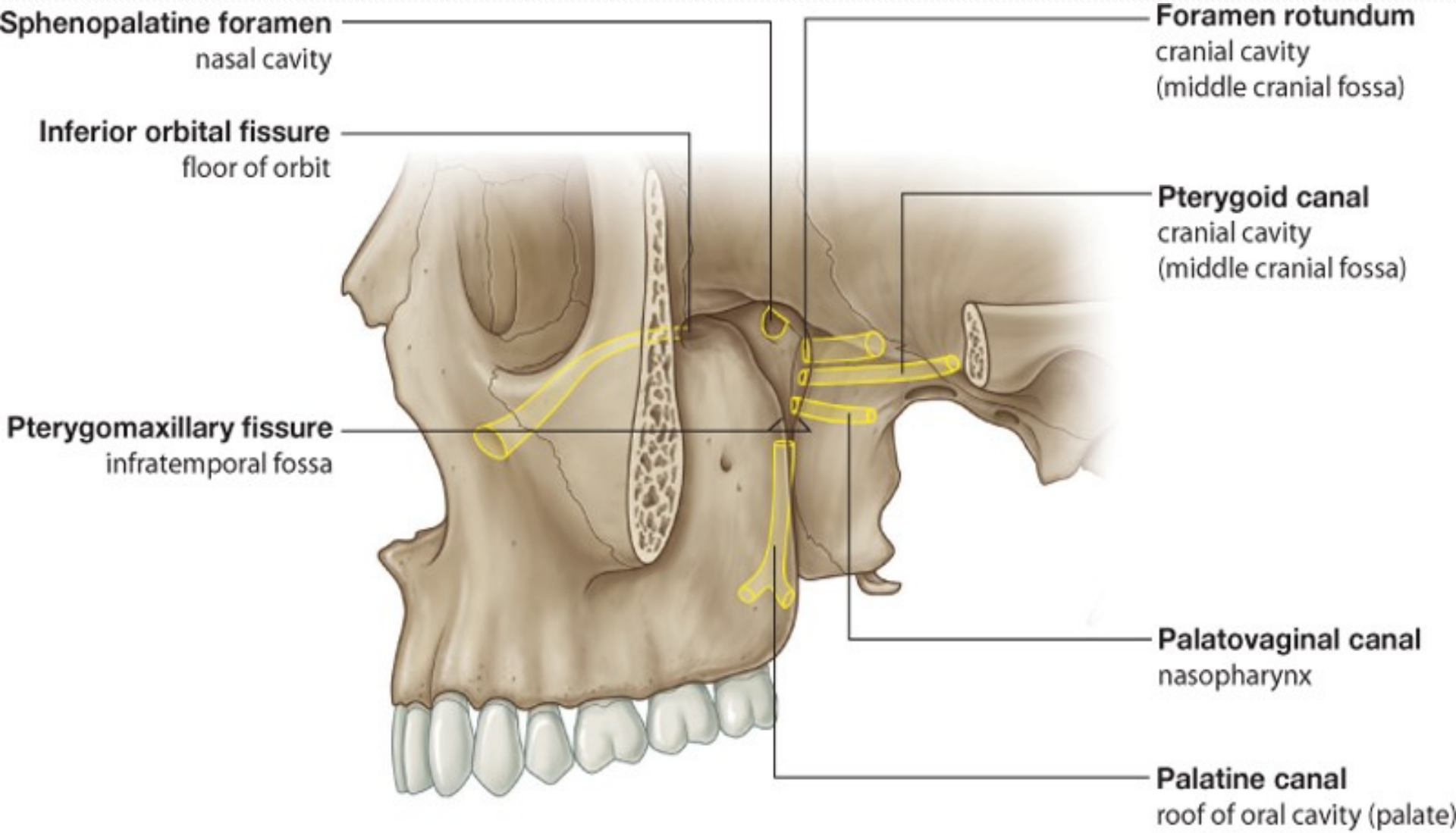


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Inferior Orbital Fissure

- The inferior orbital fissure forms the superior boundary of the pterygopalatine fossa and communicates with the **orbit**. It is a space between the sphenoid and maxilla bones.
- The **zygomatic branch** of the maxillary nerve and the **infraorbital artery and vein** pass through the inferior orbital fissure.

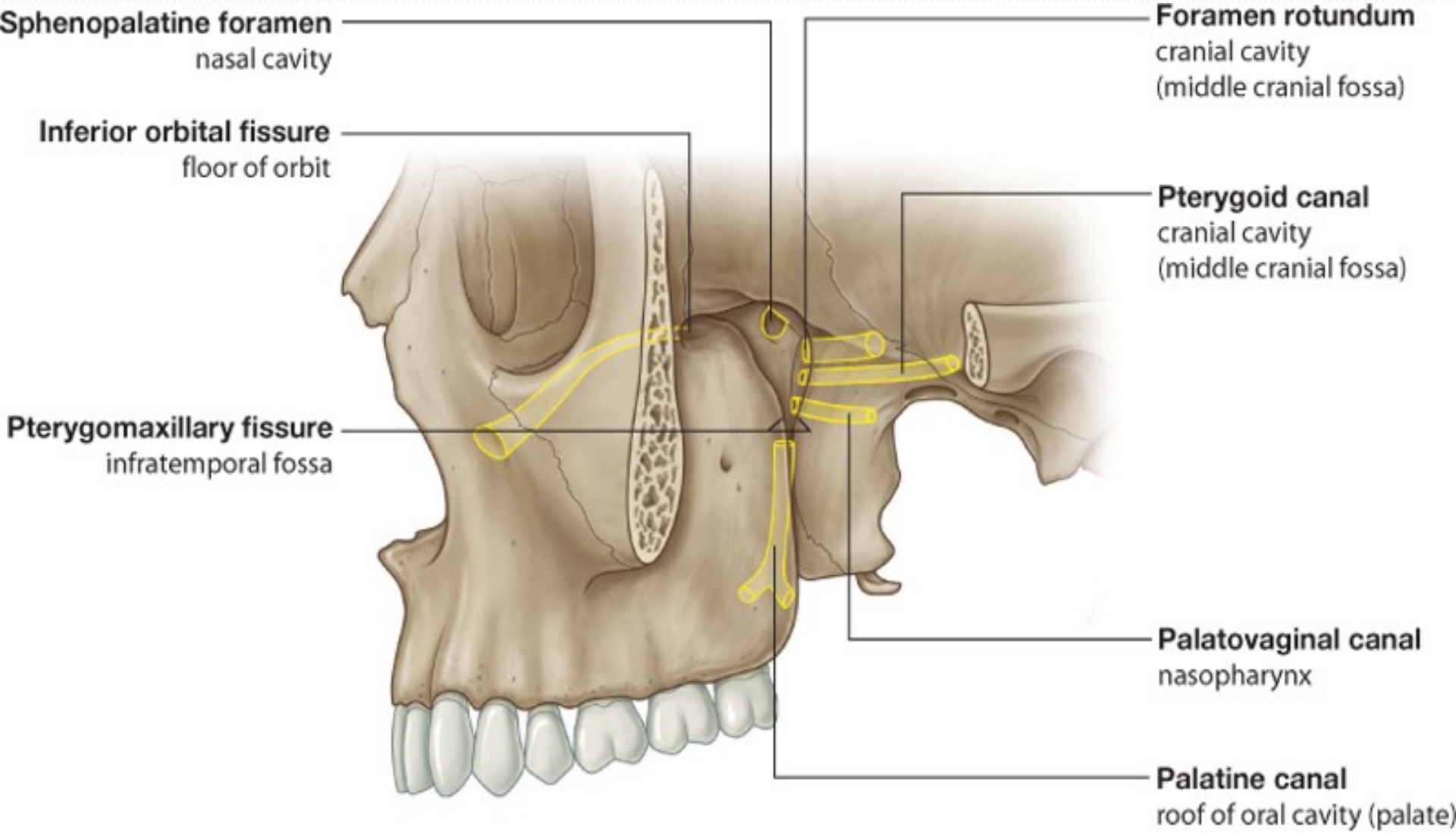


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Greater Palatine Canal

- The greater palatine canal lies in the inferior boundary of the pterygopalatine fossa, and communicates with the **oral cavity**. The canal is formed by a vertical groove in the palatine bone which is closed off by an articulation with the maxilla. Branching from the greater palatine canal are the accessory **lesser palatine canals**.
- The greater palatine canal transmits the descending palatine artery and vein, the **greater palatine nerve** and the **lesser palatine nerve**.

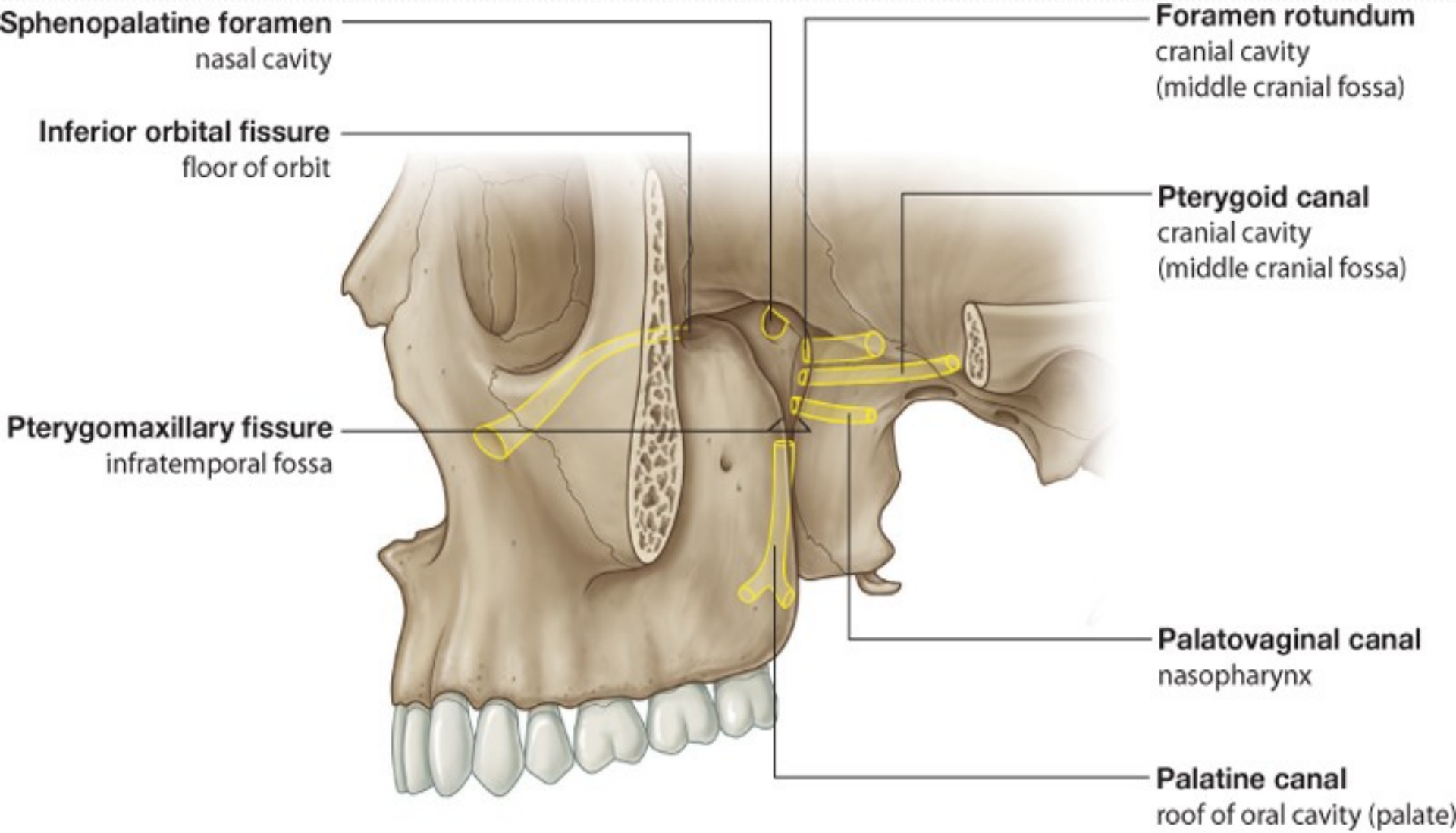


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Sphenopalatine Foramen

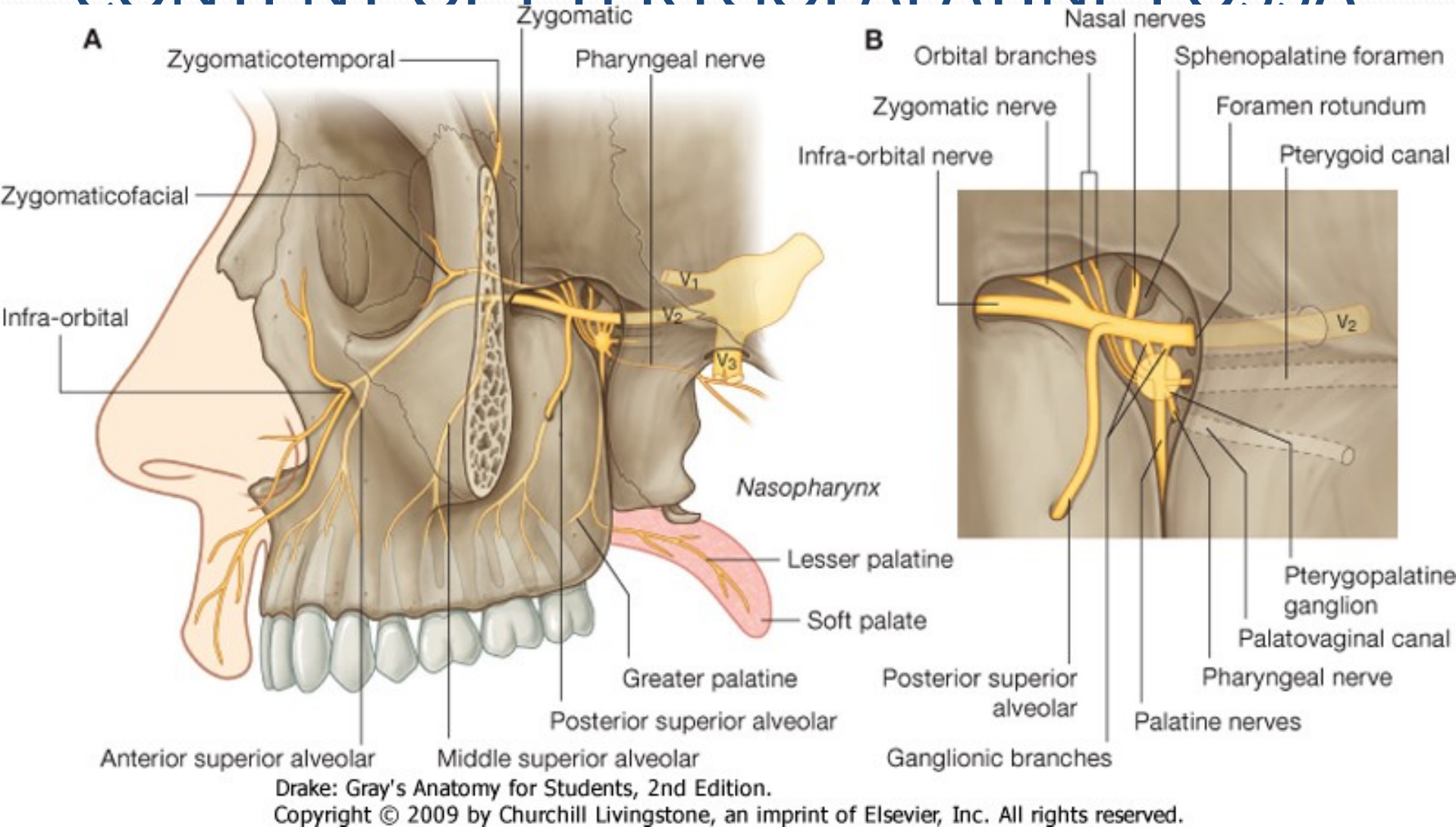
- This foramen is the only opening in the medial boundary. It connects the pterygopalatine fossa to the **nasal cavity** – specifically the superior meatus.
- It is formed by the sphenopalatine notch at the superior aspect of the perpendicular plate of the palatine bone and the body of the sphenoid.
- The sphenopalatine foramen transmits the **sphenopalatine artery and vein**, as well as the **nasopalatine nerve** (a large branch of the pterygopalatine ganglion – CNV_2).



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CONTENT OF PTERYGOPALATINE FOSSA



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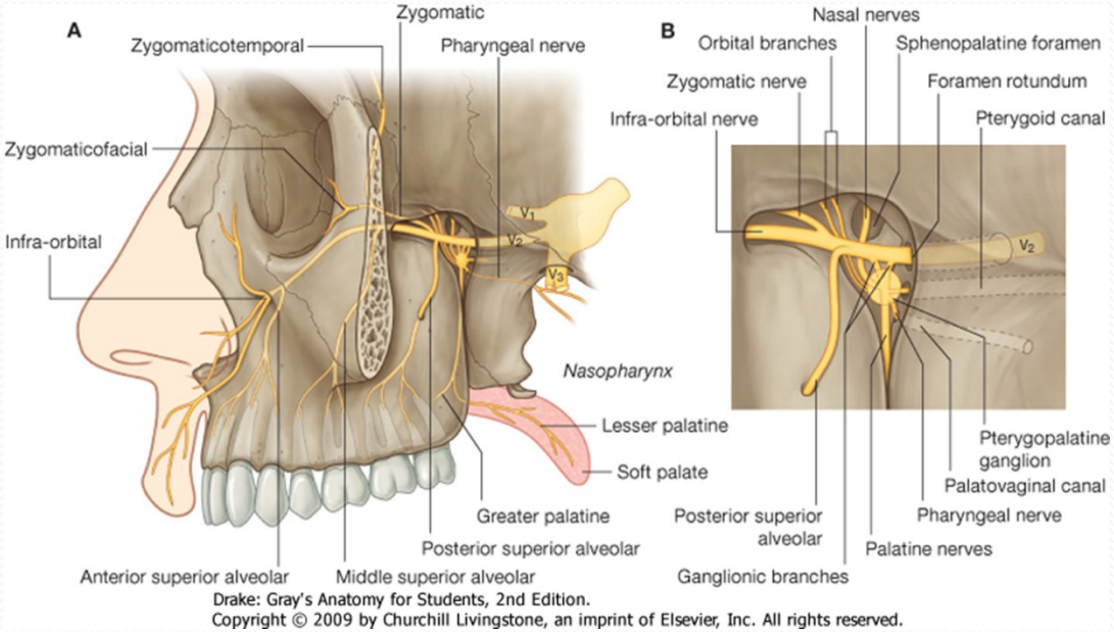
Contents

The Pterygopalatine Fossa contains many important neurovascular structures.

- the maxillary nerve and its branches,
- the pterygopalatine ganglion and
- the maxillary artery and its branches.

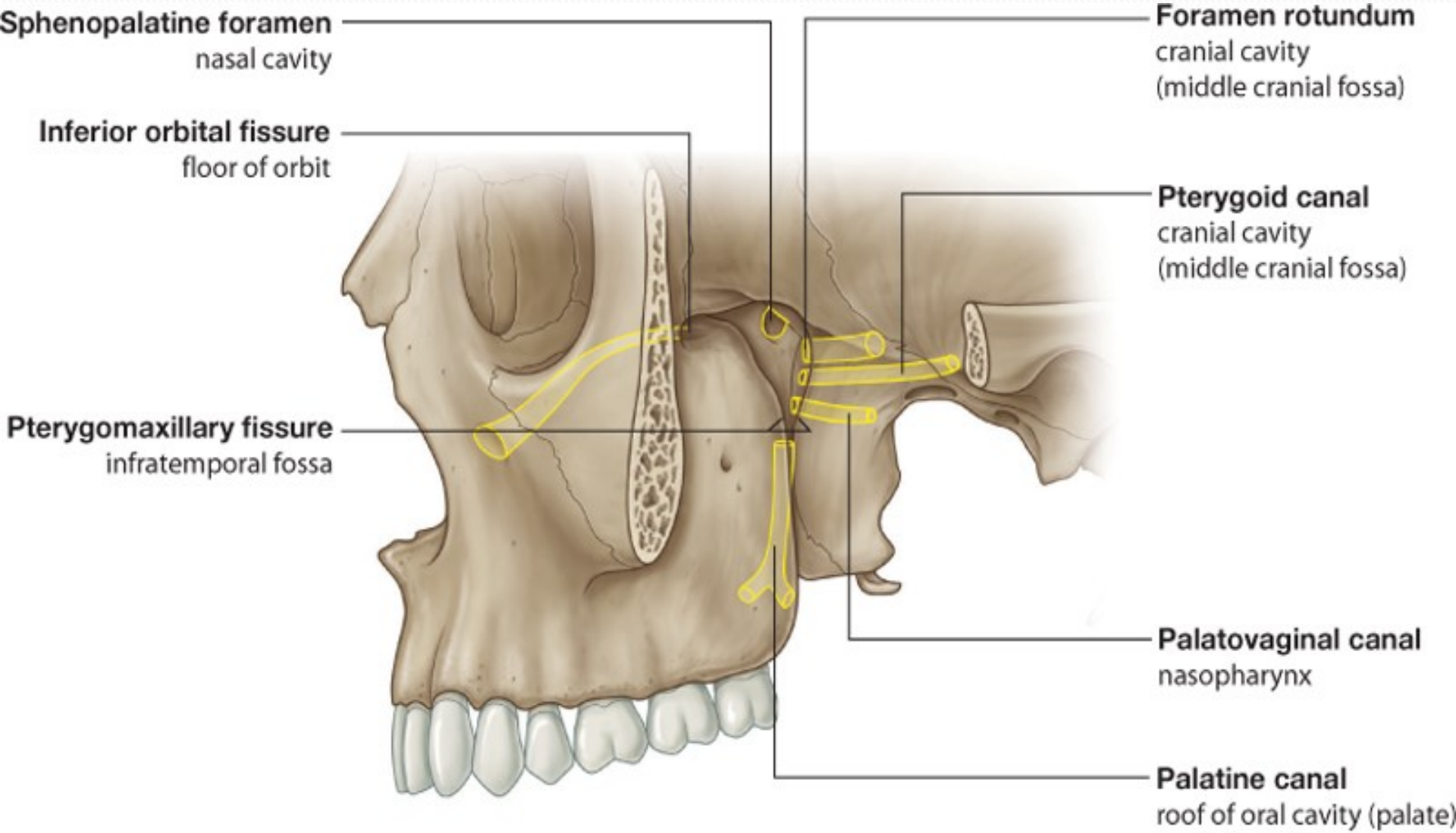
Pterygopalatine ganglion

- Largest periperal parasymp ganglion
- Relay station for secretomotor fibers to lacrimal gland, glands of nose, palate, pharyns and paranasal sinuses



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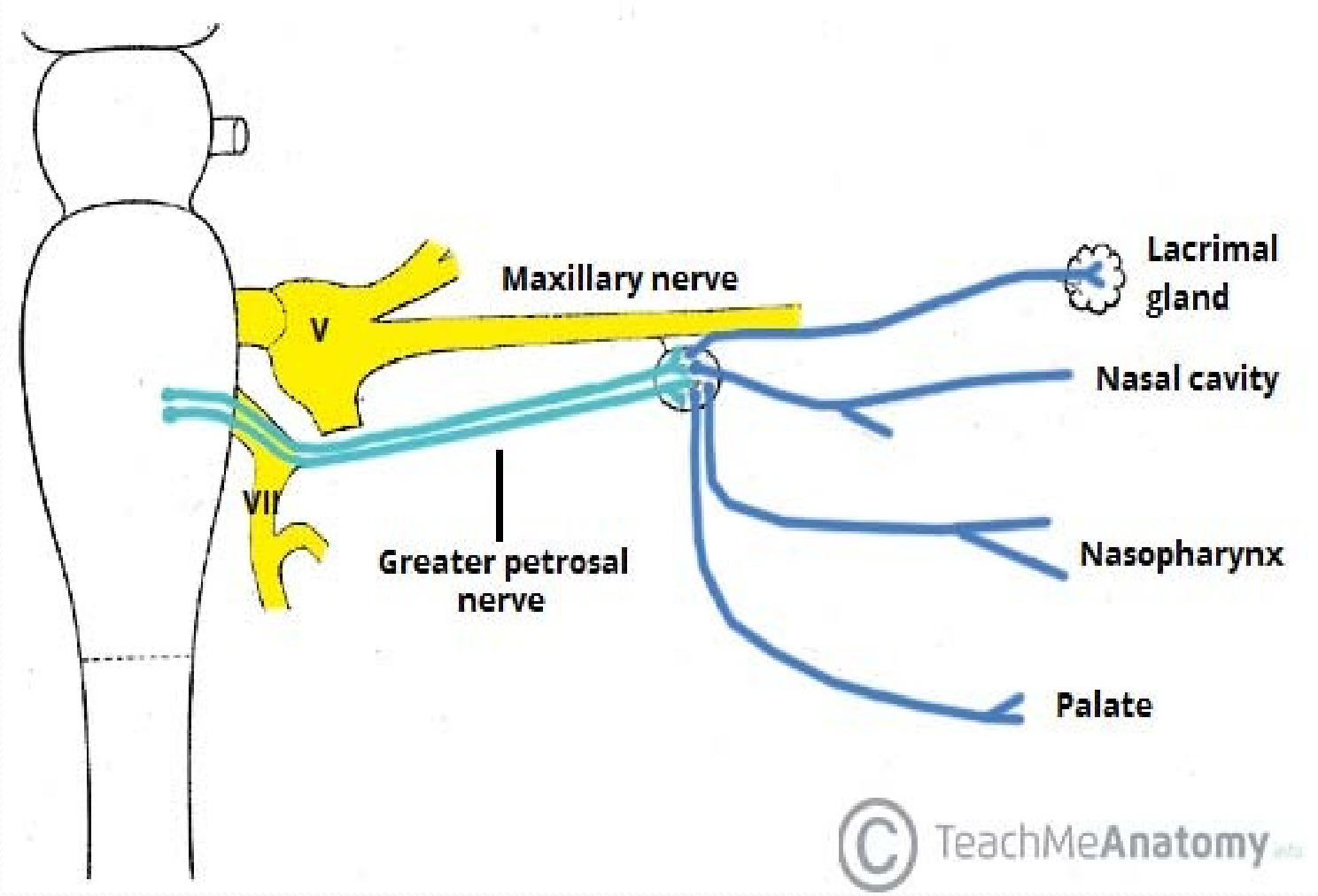
- The pterygopalatine ganglion sits deep within the pterygopalatine fossa near the sphenopalatine foramen. It is the largest parasympathetic ganglion related to branches of the maxillary nerve (via pterygopalatine branches) and is predominantly innervated by the **greater petrosal branch of the facial nerve (CNVII).**



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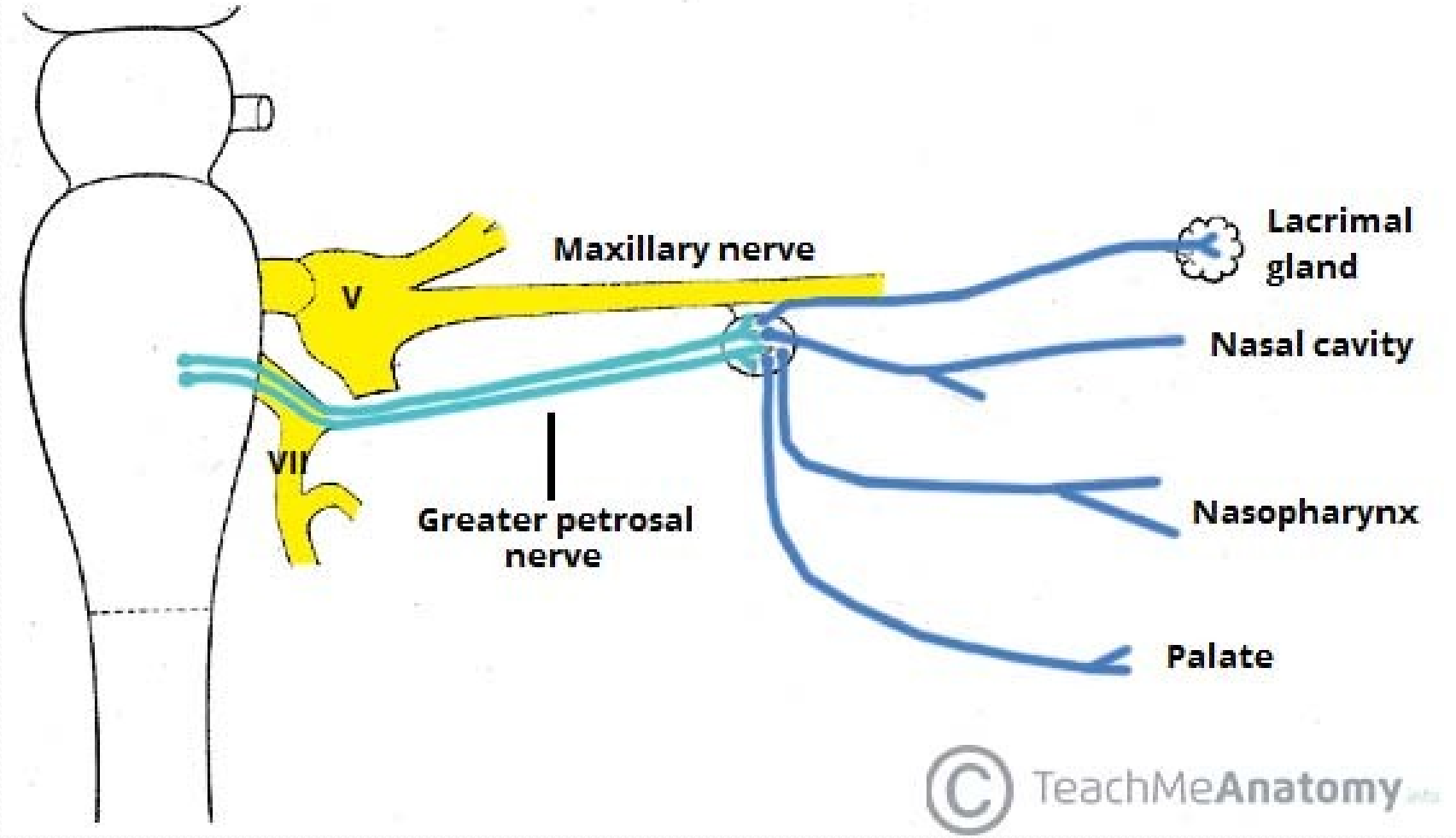
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- Postsynaptic parasympathetic fibres leave the ganglion and distribute with branches of the maxillary nerve (CNV_2). These fibres are **secretomotor** in function, and provide **parasympathetic innervation** to the lacrimal gland, and muscosal glands of the oral cavity, nose and pharynx.


Maxillary nerve
V
VII
Greater petrosal nerve
Lacrimal gland
Nasal cavity
Nasopharynx
Palate
© TeachMeAnatomy

Motor or parasympathetic root

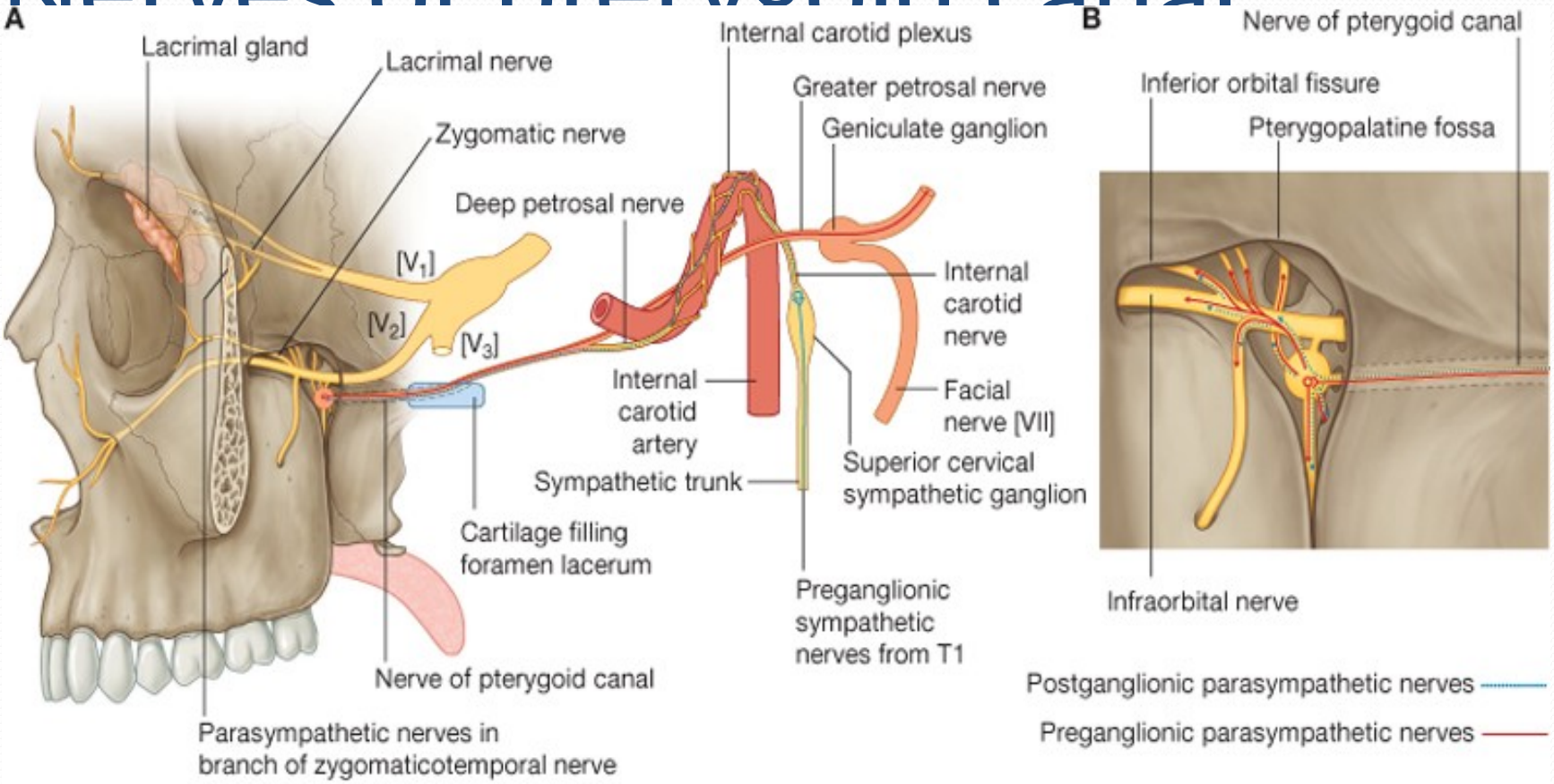
- Preganglionic fibers comes from Superior salivatory nucleusin lower pons-----pass thru nervous intermedius----fibers relay in geniculate ganglion from which arises greter petrosal nerve----which enters middle cranial fossa thru its hiatus----then it proceed towards Foramen Lacerum where it joins deep petrosal nerve to form nerve of pterygoid canal----enters pterygopalatine fossa---synapse in pterygopalatine ganglion---postganglionic fibers arise in branches of ganglion and supply lacrimal gland, glands of nose ,palate, paranasal sinuses and nasopharync


Maxillary nerve
V
VII
Greater petrosal nerve
Lacrimal gland
Nasal cavity
Nasopharynx
Palate
© TeachMeAnatomy

Sympathetic root

- Preganglionic fibers arise in T1-T4 segments of spinal cord----synapse in superior cervical ganglion----postganglionic fibers ----forms plexus around internal carotid artery----deep petrosal nerve----nerve of pterygoid canal---pass thru pterygopalatine ganglion without relay----supply vasomotor nervesto mucous membrane of nose, paranasal sinuses, palate and pharynx

Nerves of pterygoid canal

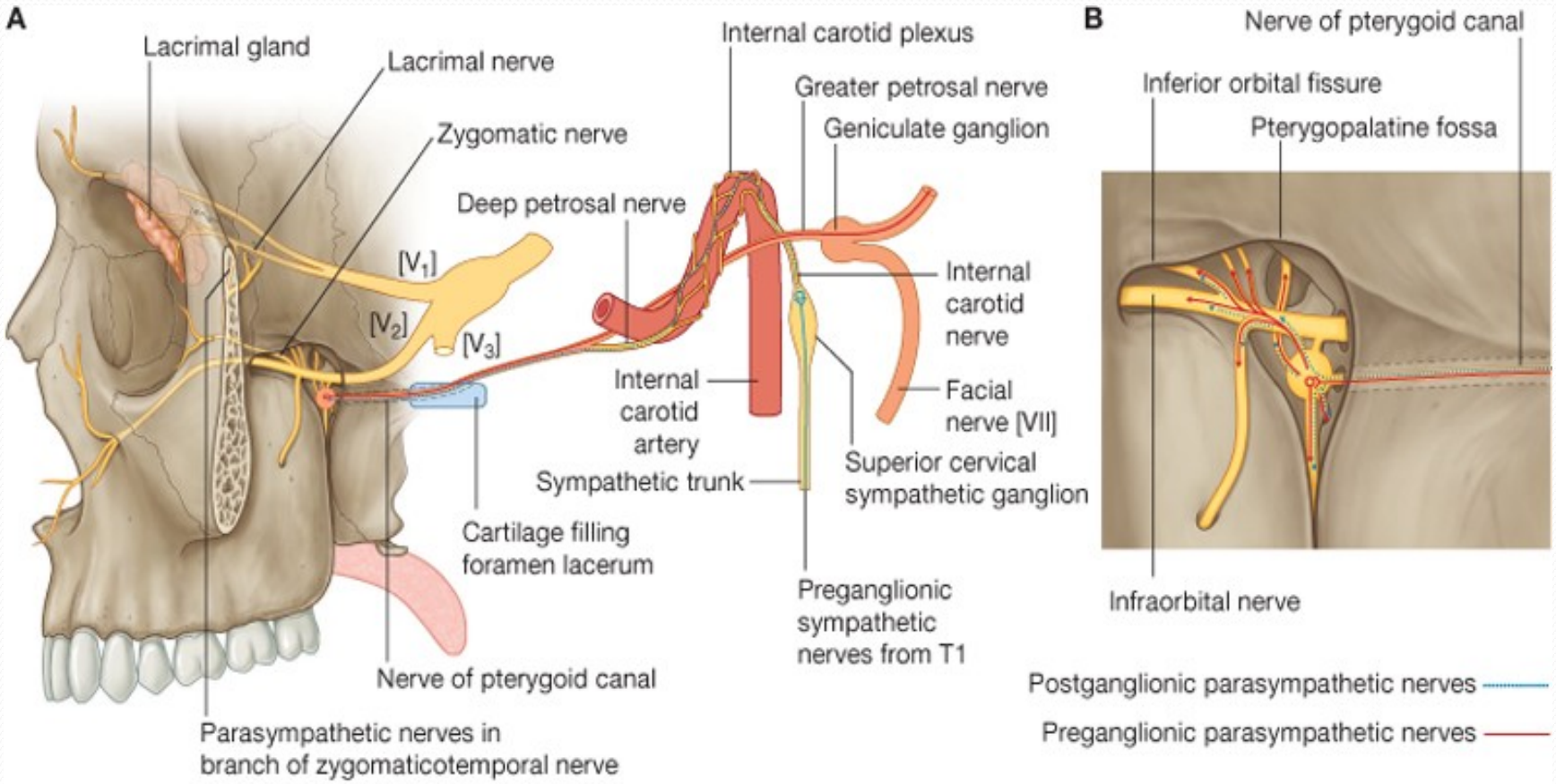


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Sensory root

- Comes from maxillary nerve----fibers pass thru pterygopalatine ganglion without relay----emerge in following barnches
- Orbital branches
- Palatine branches----greater (ant) palatine, lesser (middle and post)palatine branches
- Nasal branches---lateral post superior nasal nerve, medial post superior nasal nerve
- Pharyngeal branch

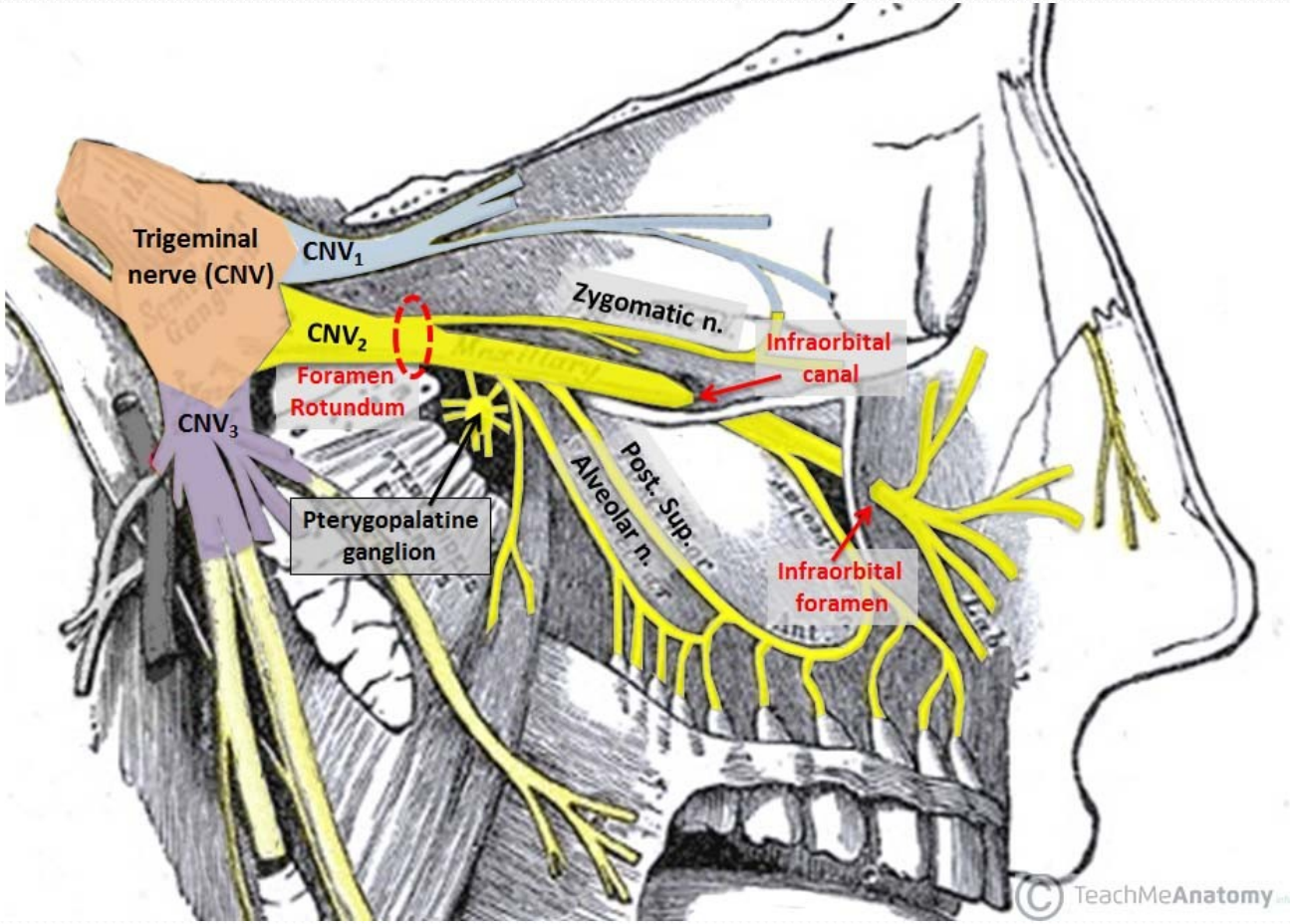
Nerves of pterygoid canal



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- The maxillary nerve is the second branch of the trigeminal nerve (CNV_2). It passes from the middle cranial fossa into the pterygopalatine fossa through the **foramen rotundum**.
- The main trunk of the maxillary nerve leaves the pterygopalatine fossa via the **infraorbital fissure**. Here, it enters the infraorbital canal of the maxilla and exits below the orbit in the infraorbital foramen to contribute to the sensory innervation of the face

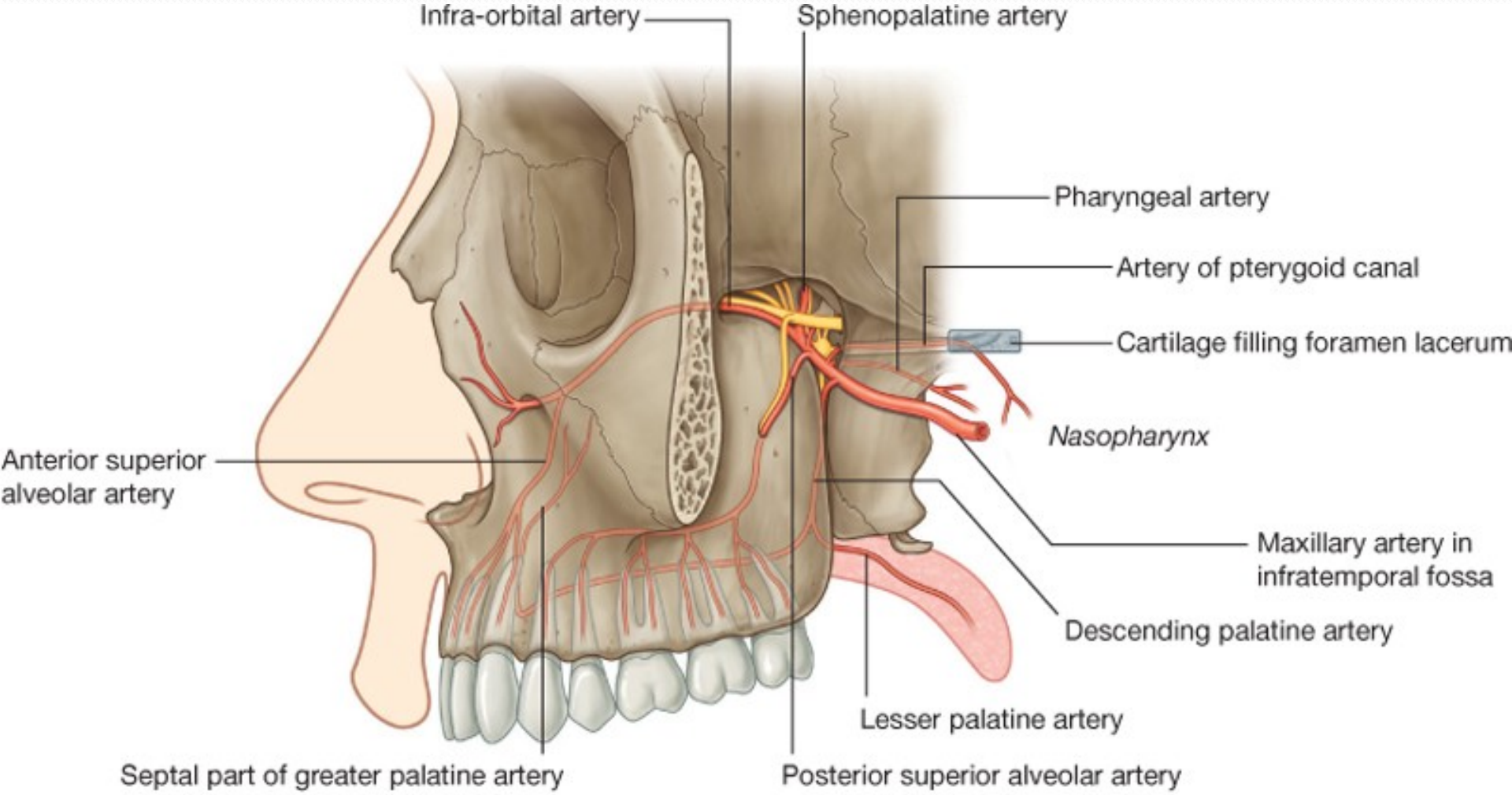
- While in the pterygopalatine fossa, the maxillary nerve gives of numerous branches including the infraorbital, zygomatic, nasopalatine, superior alveolar, pharyngeal and the greater and lesser palatine nerves. The maxillary nerve also communicates with the **pterygopalatine ganglion** (discussed below) via two small trunks, the **pterygopalatine nerves** (figure 2.1). These nerves suspend the ganglion within the pterygopalatine fossa.


Trigeminal nerve (CNV)
CNV1
CNV2
CNV3
Foramen Rotundum
Zygomatic n.
Infraorbital canal
Post. Sup. Alveolar n.
Pterygopalatine ganglion
Infraorbital foramen
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Maxillary Artery

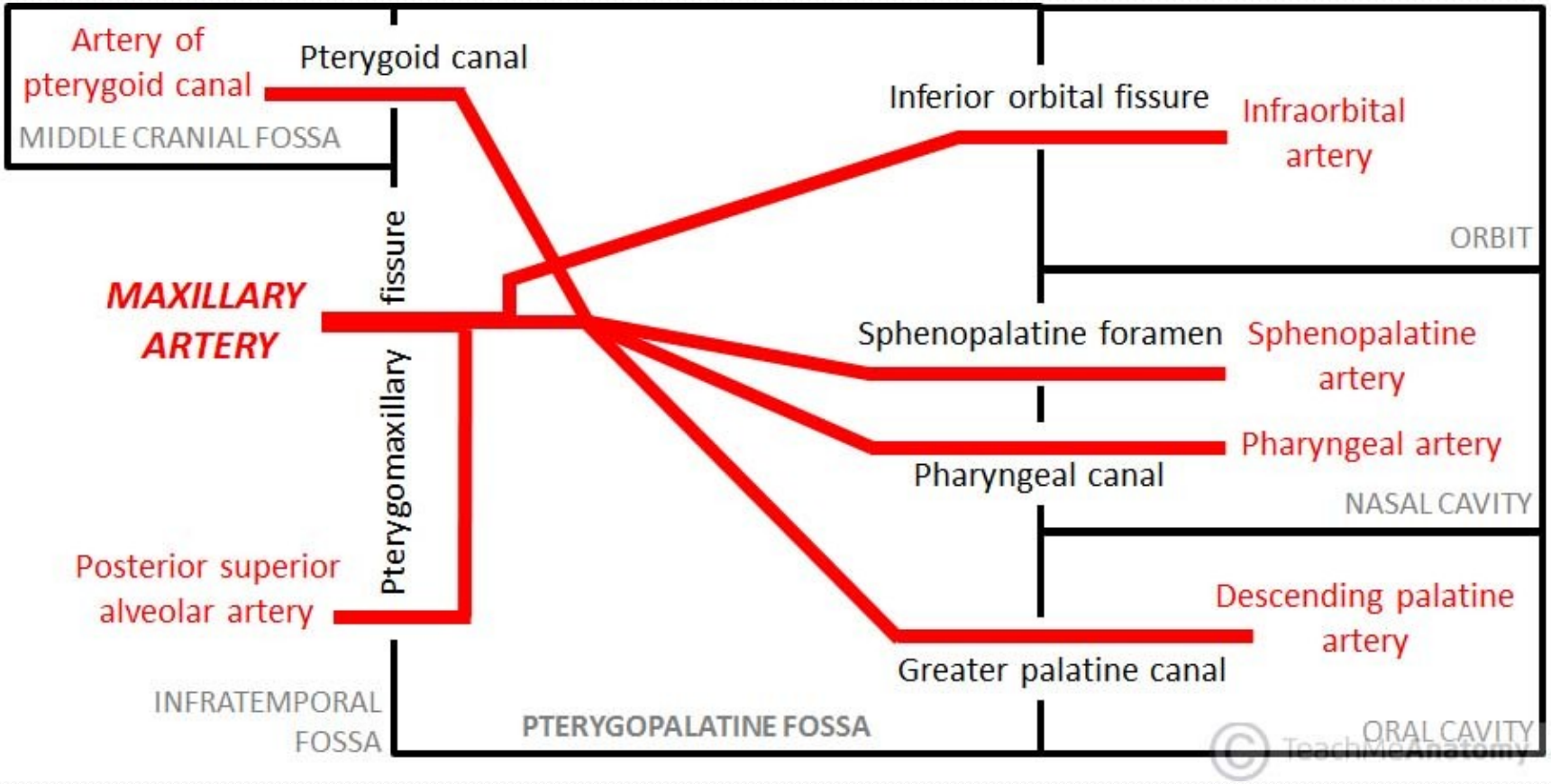
- The maxillary artery is a terminal branch of the **external carotid artery**. The terminal portion of the maxillary artery lies within the pterygopalatine fossa. Here, it separates into several branches which travel through other openings within the fossa to reach the regions they supply.

MAXILLARY ARTERY



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- These branches include:
- **Sphenopalatine artery** (to the nasal cavity).
- **Descending palatine artery** – branches into greater and lesser palatine arteries (hard and soft palates).
- **Infraorbital artery** (lacrimal gland, and some muscles of the eye).
- **Posterior superior alveolar artery** (to the teeth and gingiva).
- At their terminal ends, the sphenopalatine and greater palatine arteries anastomose at the nasal septum.


Artery of pterygoid canal
Pterygoid canal
MIDDLE CRANIAL FOSSA
fissure
MAXILLARY ARTERY
Pterygomaxillary
Posterior superior alveolar artery
INFRATEMPORAL FOSSA
PTERYGOPALATINE FOSSA
Inferior orbital fissure
Infraorbital artery
ORBIT
Sphenopalatine foramen
Sphenopalatine artery
Pharyngeal artery
Pharyngeal canal
NASAL CAVITY
Descending palatine artery
Greater palatine canal
ORAL CAVITY
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Clinical Relevance

- **Maxillary Nerve Block**
- Extensive dental surgery may require total nerve block of the maxillary branch of the trigeminal nerve (CNV_2). The maxillary nerve in the pterygopalatine fossa is most often approached intraorally via the **greater palatine canal**.

Chronic Epistaxis

- The sphenopalatine artery is often referred to as the artery of **epistaxis** (nosebleed). In cases of chronic epistaxis, the pterygopalatine fossa can be surgically approached via the maxillary sinus, and the artery ligated to control bleeding.